

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	05 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Description
FR-2	Data Integration	Merge data from multiple sources into a unified dataset.
FR-3	Data Validation	Identify and correct missing, duplicate, and inconsistent records.
FR-4	Data Transformation	Convert data into analysis-ready format using calculated fields and standard units.
FR-5	Charging Session Analysis	Analyze charging duration, energy consumed, charging speed, and charging frequency.
FR-6	Battery Performance Analysis	Evaluate battery capacity, State of Charge (SoC), State of Health (SoH), and charging efficiency.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be intuitive and easy to use for all users.
NFR-2	Scalability	Handle increasing EV datasets without significant performance degradation.
NFR-3	Reliability	Provide accurate and consistent analytical results.
NFR-4	Security	Protect user and EV data through authentication and access control.